Methodology: Counterfactuals

The housebuilding crisis: The UK's 4 million missing homes

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Corrected edition, September 2026

 This corrected edition revises the equations and worked figures so that they match the code that reproduces the report's results: `code/Counterfactual Replication.R` for the counterfactuals and `code/Summary Replication.R` for the housing stock estimate. Every worked figure has been recomputed from `data/Combined.csv`. The final section lists the changes from the February 2023 text.

Introduction

In the report **The housebuilding crisis: The UK's 4 million missing homes**, housing outcomes are defined as the number of homes per (thousand) people. This is a simple metric for general comparisons in housing quality and affordability.

One of the goals of the report was to calculate exactly how much underbuilding had taken place in the UK since 1955 compared to other countries, and how this affected outcomes in the UK.

Comparisons that simply applied the housebuilding rates of other countries to the UK generated unbuilt backlogs that were too large, as other countries in 1955 had worse outcomes and higher population growth, and therefore needed to build more.

More reasonable estimates could be produced by calculating the housebuilding required to reach the number of homes per person that other European countries had in 2015, but these were still unsatisfactory, for two reasons.

First, investigating differences in housebuilding by tenure was not possible with this method, despite being a key aspect of the report.

And second, inconsistencies in the data between countries make it difficult to draw comparisons based on the number of dwellings recorded in 2015. Both the UK, and wherever possible the UN data, record the total housing stock until 2000. However, the statistical agencies of many other countries today only record primary residences, and omit second homes and vacant stock. Therefore, using the publicly available 2015 dwellings per person ratios from national statistics agencies to estimate UK underbuilding from 1955 generates underestimates and inconsistent results between countries.

To solve this, Centre for Cities used a new method that combined both approaches. The report's estimates of the total amount of underbuilding in the UK from 1955 to 2015 relative to other countries are the difference between the number of homes the UK would have had if it had built at the rate of another country and the number it actually had, controlling for the difference in the initial stock of homes per person and for different population growth rates.

The number of homes per person in any year is

$$\text{Homes per person}_t = \frac{\text{Previous year's housing stock} + \text{Net additions}_t}{\text{Population}_t},$$

with

$$\text{Net additions}_t = \text{Homes built}_t - \text{Homes demolished}_t.$$

This calculation is done twice: first to estimate the actual British net additions every year, and then again for each "counterfactual" version of the UK, adjusted for each country comparison (e.g. the Netherlands). The difference between the final values for the actual UK and each counterfactual is the estimate of the housebuilding that would have taken place in the UK had it followed the housing policy approach of that country.

Part I explains the calculation for the actual UK, including how many homes the private and public sectors added from 1955 to 2015. Part II explains how it is adjusted for each counterfactual.

Data and notation

Years are indexed $t = 1, \ldots, T$ for 1955 to 2015, so $T = 61$. Worked figures are in thousands of homes unless stated. They are rounded for display only: the code rounds nothing until its final output, so displayed figures may not add exactly. For each country the data provide:

- S_t, the estimated housing stock for year t (`X5 (Estimate)`), constructed as described under *Interpolation*;
- B_t, gross completions (`Y7`);
- θ_t^{Priv} and θ_t^{Pub}, the private and public shares of completions (columns `4` and `1`);
- population in millions, written P_t for the UK and Q_t for a comparator country, where P_0 and Q_0 are the 1954 values;
- homes per 1,000 people in 1955, written PC_1 for the UK and QK_1 for a comparator.

Building rates are shares of the estimated stock, by tenure:

$$b_t^{\text{Priv}} = \frac{\theta_t^{\text{Priv}} B_t}{S_t}, \qquad b_t^{\text{Pub}} = \frac{\theta_t^{\text{Pub}} B_t}{S_t}, \qquad b_t = b_t^{\text{Priv}} + b_t^{\text{Pub}},$$

for the UK, and c_t^{Priv}, c_t^{Pub} and c_t are defined in the same way for a comparator. The two tenure shares are used as recorded and do not always sum to one: they fall to as little as 0.37 for Finland in 1955–79, and range from 0.84 to 1.05 for Switzerland in 1955–78. In those years the building counted by the method differs from recorded completions.

The UK demolition rate is $d_t = E_t/S_t$, where E_t is reported demolitions. In the 10 years without a report, E_t is completions multiplied by the ratio of demolitions to completions, interpolated across gaps of up to two years and otherwise carried forward from the most recent reported year.

Part I – Calculating the actual British values

Four steps are required:

1. The housing stock for each year from 1955 to 2015.
2. The net change in housing stock between 1955 and 2015.
3. Gross housebuilding and gross demolitions from 1955 to 2015.
4. Net additions by tenure from 1955 to 2015.

The UK is put through exactly the same calculation as each counterfactual, so that the two are consistent. The resulting stock is a model value rather than the national statistic: for 2015 it is 27.65 million against an estimated 28.28 million, 2.2 per cent lower.

Stage 1: The housing stock for each year from 1955 to 2015

The calculation starts from the estimated stock in 1955, $H_0 = S_1 = 15{,}418$. Each year the stock grows by that year's building rate net of its demolition rate:

$$H_t = H_{t-1}(1 + b_t - d_t), \qquad \text{so} \qquad H_t = H_0 \prod_{s=1}^{t} (1 + b_s - d_s).$$

In 1955 and 1956 the rates are $b_1 = 2.13\%$, $d_1 = 0.36\%$, $b_2 = 1.98\%$ and $d_2 = 0.41\%$, which give $H_1 = 15.69$ million and $H_2 = 15.94$ million. Continuing to 2015 gives $H_T = 27.65$ million, and dividing by population gives the number of homes per person in every year.

Stage 2: The net change in housing stock between 1955 and 2015

The net change is the difference between the final and initial stock, which is the sum of each year's net additions:

$$\Delta H = H_T - H_0 = \sum_{t=1}^{T} H_{t-1}(b_t - d_t).$$

For the UK, $\Delta H = 27{,}651.0 - 15{,}418.0 = 12{,}233.0$.

Stage 3: Gross housebuilding and gross demolitions

The net change in total dwellings is not enough to estimate additions by tenure: gross housebuilding and gross demolitions are needed. The code measures both against the stock at the *end* of each year, H_t, and splits each year's demolitions by that year's tenure shares:

$$G^{\text{Priv}} = \sum_{t=1}^{T} H_t \, b_t^{\text{Priv}}, \qquad E^{\text{Priv}} = \sum_{t=1}^{T} H_t \, \theta_t^{\text{Priv}} d_t, \qquad N^{\text{Priv}} = G^{\text{Priv}} - E^{\text{Priv}},$$

and likewise for public building, with totals $G = G^{\text{Priv}} + G^{\text{Pub}}$, $E = E^{\text{Priv}} + E^{\text{Pub}}$ and $N = N^{\text{Priv}} + N^{\text{Pub}}$. For the UK,

$$N = G - E = 15{,}512.1 - 3{,}151.2 = 12{,}360.9.$$

Because each year's rates are applied to the end-of-year stock rather than the start-of-year stock that generated them, N exceeds the net change in the stock by

$$\varepsilon = N - \Delta H = 12{,}360.9 - 12{,}233.0 = 127.9.$$

When the tenure shares sum to one, this is exactly $\varepsilon = \sum_t H_{t-1}(b_t - d_t)^2$, a second-order term.

Stage 4: Net additions by tenure

Splitting net change in stock by tenure requires further assumptions about demolitions (and, for the counterfactuals, population growth). Demolitions are allocated each year to **the same tenure proportion as the homes built in that year**; in Part II, the population and initial-stock adjustments are allocated by the comparator's tenure mix of gross building.

British data is available from 1955 to 2015 for gross housebuilding by tenure and gross demolitions, but not the necessary data to measure demolitions by tenure. Although for the latter part of the period some data does exist on demolitions by tenure of final occupier for England, data does not exist on demolitions by tenure of *construction*. As this report is concerned with the contribution of the private and public sector to total net additions of stock, it is the latter we are interested in.

To take an example – consider a local authority which built ten council houses, and four of them were sold to tenants under Right to Buy. One remaining council tenancy property and a Right to Buy property were both then demolished. The net change in tenure by occupation is an additional five council dwellings and three homeowner dwellings. But when considering the net change by construction, the public sector added eight dwellings, despite Right to Buy and the demolition of an owner-occupied dwelling.

As data on demolitions by tenure is tracked by final occupier, not by construction, this presents a problem for the methodology. Former council homes sold under Right to Buy and former private properties sold to Registered Providers will have their demolitions recorded incorrectly. The report must make an assumption about demolitions for the entire period.

Demolitions were higher in the postwar period (specifically the 1960s) and concentrated on private sector stock in slum clearances. Demolitions since 1980 were lower over a longer time period, and primarily were of social housing and private properties often built by the public sector (e.g. leaseholders on council estate regenerations).

Allocating each year's demolitions to that year's tenure mix puts 58.6 per cent of UK demolitions from 1955 to 2015 on private-built homes. This is below the private share of gross building (64.4 per cent) because demolition was heaviest, relative to building, in years when public building made up a larger share. The individual results in each year are skewed, but the cumulative results are satisfactory.

Finally, the tenure totals are scaled so that they add up to the net change in the stock, removing ε in proportion to each tenure's net building:

$$A^{\text{Priv}} = N^{\text{Priv}} \frac{\Delta H}{N}, \qquad A^{\text{Pub}} = N^{\text{Pub}} \frac{\Delta H}{N}, \qquad A^{\text{Priv}} + A^{\text{Pub}} = \Delta H.$$

UK, 1955–2015 (thousands)	Private	Public	Total
Gross building, G	9,985.5	5,526.7	15,512.1
Demolitions, E	1,847.7	1,303.5	3,151.2
Net building, N	8,137.7	4,223.2	12,360.9
Net additions, A	8,053.5	4,179.5	12,233.0

The UK therefore added 8.05 million private and 4.18 million public homes from 1955 to 2015, a split of 66:34.

Part II – Calculating counterfactual values for British housebuilding

Having calculated the required figures for Britain, we now move to the method used to make credible adjustments for other countries.

The central estimate we are making is **how many extra homes the UK would have had in 2015 if it had built at the rate of another Western European country from 1955.** As accurate values for the net change in housing stocks both in general and by tenure are not available, this is calculated using each country's gross building and UK demolitions.

However, simply repeating the British method with another country's building rates will not produce a credible result. This is because:

Other countries have different population growth rates compared to Britain. If the population growth rate is different to the UK's, then the same gross increase will not produce the same change in the number of homes per person. As all countries except Austria and Belgium had higher population growth than the UK, adjusting for this makes the average estimate more conservative.

Other countries began the period with a different per person stock of homes compared to Britain. The average western European country began the period with a smaller stock of homes per person than the UK, which means some of their housebuilding after 1955 will have been to catch up to a housing stock that the UK had already built before 1955. Adjusting for this makes the estimate more conservative.

Therefore, these effects need to be controlled for when calculating the counterfactual value of the United Kingdom, so that it has:

- the same initial housing stock to population ratio as the counterfactual country;
- the same population growth rate as the counterfactual country.

Methodology

A counterfactual UK that followed a comparator country (e.g. the Netherlands) is built in the same way as the actual UK in Part I, with three adjustments:

1. its initial stock gives it the comparator's 1955 ratio of homes per person;
2. its stock is rescaled each year by the difference between UK and comparator population growth;
3. it builds at the comparator's rates, and demolishes at the UK's rate adjusted for how many more or fewer homes per person it has than the actual UK.

The counterfactual's annual stock is then used for the same tenure calculation as in Part I:

4. gross building, demolitions and net building by tenure;
5. the discrepancy that the population rescaling creates between net building and the change in the stock.

Finally:

6. the difference between the counterfactual and the actual UK is split by tenure.

Stage 1: Adjust the counterfactual's initial stock of homes per person

The counterfactual starts with the UK's 1955 population housed at the comparator's 1955 ratio of homes per person:

$$\sigma_0 = P_1\, QK_1 = H_0\, \frac{QK_1}{PC_1},$$

where the second form uses $PC_1 = H_0/P_1$. In 1955 the Netherlands had 233.69 dwellings per 1,000 people against the UK's 302.49, and the UK's population was 50.97 million. So the initial stock of the counterfactual UK based upon the Netherlands is

$$\sigma_{\text{Neth}} = 50.97 \times 233.693 = 11{,}911.3.$$

Stage 2: Net change in stock for the counterfactual UK

The counterfactual builds at the comparator's rate c_t and demolishes at a rate D_t, which is set out in Stage 4. Before any adjustment for population, its stock would grow each year by the factor $1 + c_t - D_t$: in the first year, from σ_0 to $\sigma_0(1 + c_1 - D_1)$.

Stage 3: Adjust net change in stock for population growth

Population growth differs between countries every year. This means that an identical increase in the overall housing stock will not result in an identical increase in the stock of homes per person. Places with lower population growth do not need to build as many homes to achieve the same improvements to housing outcomes. Therefore, the change in population relative to Britain is applied to the housing stock every year.

The code measures each year's population growth relative to that year's population:

$$\Delta P_t = 1 + \frac{P_t - P_{t-1}}{P_t}, \qquad \Delta Q_t = 1 + \frac{Q_t - Q_{t-1}}{Q_t},$$

a close approximation to the growth factors P_t/P_{t-1} and Q_t/Q_{t-1}. Writing $R_t = \Delta P_t/\Delta Q_t$, the counterfactual stock ν_t is

$$\nu_t = \nu_{t-1}(1 + c_t - D_t)\, R_t, \quad \nu_0 = \sigma_0, \qquad \text{so} \qquad \nu_t = \sigma_0 \prod_{s=1}^{t} (1 + c_s - D_s)\, R_s.$$

Multiplying by R_t means that the counterfactual's homes per person grow as they would with the comparator's building and population growth, while its stock is always expressed for the UK's population. The Netherlands grew faster than the UK: in 1955, $R_1 = 0.9917$, or about $1/1.008$, so the counterfactual stock is scaled down slightly. For the UK itself, $R_t = 1$ and $D_t = d_t$, so $\nu_t = H_t$.

Stage 4: Adjust demolition rates for the counterfactual's stock of homes

Accurate and consistent demolition data does not exist for European countries apart from Switzerland. For all other countries, the demolition rate is set through a two-step process.

Initially, given that this is a model of Britain's housing, the demolition rates are assumed to be identical to Britain's, as it is mostly pre-1947 British stock which is being demolished. This gives a first-round stock ξ_t:

$$\xi_t = \xi_{t-1}(1 + c_t - d_t)\, R_t, \qquad \xi_0 = \sigma_0.$$

However, the UK's demolition rate needs to be adjusted for two reasons. First, the UK's demolitions are concentrated in the 1960s due to a policy choice, with little demolition earlier or after that. Applying this to the counterfactual version of the UK building at another rate will have an inconsistent effect on the net change in dwellings. Second, as the main target of demolitions was unfit pre-modern housing, it is reasonable to assume that demolition rates would be in proportion to the stock of housing. If the UK had had less housing than it actually did in 1955, then it likely would have demolished fewer dwellings in the years after.

Because ξ_t and H_t are both stocks for the UK's population, ξ_t/H_t is the ratio of the counterfactual's homes per person to the actual UK's. The demolition rate is moved by one hundredth of the proportional gap:

$$D_t = d_t + \frac{1}{100}\left(\frac{\xi_t}{H_t} - 1\right).$$

A counterfactual with 1 per cent more homes per person than the actual UK demolishes at a rate 0.01 percentage points higher; one with 1 per cent fewer, at a rate 0.01 percentage points lower. In 1955 the Netherlands counterfactual has $\xi_1/H_1 = 0.768$, 23.2 per cent fewer homes per person than the UK, so its demolition rate is

$$D_1 = 0.357\% - 0.232\% = 0.125\%.$$

The effect of this over time smooths out demolitions from 1955 to 2015. In 1955 the actual UK has more homes per person than six of the eleven comparators, and by 2015 every counterfactual has more homes than the actual UK. So most counterfactual versions of the UK have fewer demolitions at the beginning of the period (and in the 1960s), when they have fewer homes per person than the UK, and more demolitions towards the end of the period, when they have overtaken the UK and actual UK demolitions have dropped to very low levels.

Nothing stops D_t falling below zero. It does so for the Ireland counterfactual in 18 years (1978 to 1996), where the negative demolitions add about 239,000 homes, and for the Netherlands in 1992, where the effect is negligible. The script also builds a revised table in which Ireland and Sweden use their own reported demolitions; see the Notes.

Switzerland is the exception: its demolition data are of high enough quality to use directly. Its D_t is its own demolitions divided by its own estimated stock, with gaps of up to two years interpolated and longer gaps filled using the ratio of demolitions to completions from the most recent reported year (or the first, for years before any report).

The net result is conservative: across the eleven comparators, it reduces the average number of additional homes from 4.87 million, with $D_t = d_t$, to 4.25 million.

Stage 5: Gross building, demolitions and net building by tenure

Part I is now applied to the counterfactual stock. As there, building and demolitions in year t are measured against the end-of-year stock ν_t, and demolitions are split by the comparator's tenure shares:

$$J_B^{\text{Priv}} = \sum_{t=1}^{T} \nu_t\, c_t^{\text{Priv}}, \qquad J_D^{\text{Priv}} = \sum_{t=1}^{T} \nu_t\, \theta_t^{\text{Priv}} D_t, \qquad J_N^{\text{Priv}} = J_B^{\text{Priv}} - J_D^{\text{Priv}},$$

and likewise for public building, with totals J_B, J_D and J_N. For the Netherlands counterfactual:

Netherlands counterfactual (thousands)	Private	Public	Total
Gross building, J_B	14,201.1	9,881.8	24,083.0
Demolitions, J_D	952.5	769.6	1,722.2
Net building, J_N	13,248.6	9,112.2	22,360.8

Stage 6: Adjust for the population discrepancy

Net building should equal the change in the stock, but the population rescaling in Stage 3 changes the stock without any building or demolition. The discrepancy is

$$\Gamma = J_N - (\nu_T - \sigma_0) = \sum_{t=1}^{T} \gamma_t, \qquad \gamma_t = \nu_t\big(c_t - (\theta_t^{\text{Priv}} + \theta_t^{\text{Pub}})D_t\big) - (\nu_t - \nu_{t-1}).$$

When the tenure shares sum to one,

$$\gamma_t = \nu_{t-1}\big[1 - R_t\big(1 - (c_t - D_t)^2\big)\big],$$

which combines the population rescaling with the same end-of-year timing term as ε in Part I. For the UK, $R_t = 1$ and $\Gamma = \varepsilon$. Due to compounding, the discrepancy runs into millions of homes in the counterfactuals from 1955 to 2015.

For the Netherlands counterfactual, $\nu_T = 30{,}486.7$, so

$$\nu_T - \sigma_0 = 30{,}486.7 - 11{,}911.3 = 18{,}575.4, \qquad \Gamma = 22{,}360.8 - 18{,}575.4 = 3{,}785.4.$$

Stage 7: Compare to the UK, by tenure

The aim of the counterfactuals is **how many extra homes Britain would have of each tenure had it built at the rate of another European country.** The total is the difference in final stocks, which already reflects the initial difference in homes per person, population growth and demolitions:

$$\text{Total surplus} = \nu_T - H_T.$$

For the Netherlands, Total surplus = 30,486.7 – 27,651.0 = 2,835.7.

Part of that difference is a difference in net building, $J_N - N$. The rest is

$$\Delta = (\nu_T - H_T) - (J_N - N) = (\sigma_0 - H_0) - (\Gamma - \varepsilon) :$$

the initial gap in stock, less the comparator's population discrepancy net of the UK's own timing term. It is allocated between tenures by the comparator's private share of gross building, $s = J_B^{\text{Priv}}/J_B$:

$$\begin{aligned} \text{Private surplus} &= \big(J_N^{\text{Priv}} - N^{\text{Priv}}\big) + s\,\Delta, \\ \text{Public surplus} &= \big(J_N^{\text{Pub}} - N^{\text{Pub}}\big) + (1-s)\,\Delta. \end{aligned}$$

The two add up to the total surplus. Country surpluses are measured against the UK's net building N^{Priv} and N^{Pub} before the scaling in Part I, Stage 4; the UK's timing term ε enters through Δ instead.

For the Netherlands, $s = 0.58968$ and

$$\Delta = (11{,}911.3 - 15{,}418.0) - (3{,}785.4 - 127.9) = -3{,}506.7 - 3{,}657.5 = -7{,}164.2,$$

so

$$\text{Private surplus} = (13{,}248.6 - 8{,}137.7) + 0.58968 \times (-7{,}164.2) = 5{,}110.9 - 4{,}224.6 = 886.3,$$
$$\text{Public surplus} = (9{,}112.2 - 4{,}223.2) + 0.41032 \times (-7{,}164.2) = 4{,}889.0 - 2{,}939.6 = 1{,}949.4.$$

The results table reports these to four significant figures: for the Netherlands, 886,300 private, 1,949,000 public and 2,836,000 in total, with a private:public mix of gross building of 59:41. The United Kingdom row reports A^{Priv}, A^{Pub} and ΔH from Part I (8,054,000, 4,179,000 and 12,230,000, a mix of 66:34, from N^{Priv}/N). The Western European Average row is the simple mean of the eleven comparators: 5,301,000 private, −1,047,000 public and 4,254,000 in total.

Notes

No floor on public additions. The code applies no floor, and none binds: the counterfactual's public net additions, $J_N^{\text{Pub}} + (1 - s)\Delta$, are positive for every comparator. The smallest is Switzerland's, at 401,600.

The published Table 3. The report's Table 3 has the same total on every row as this calculation but a different private/public split. Its UK row, 7,875,000 private and 4,358,000 public, splits total UK demolitions by the cumulative private share of gross building instead of year by year:

$$G^{\text{Priv}} - (G - \Delta H)\,\frac{G^{\text{Priv}}}{G} = 7{,}875.$$

That calculation is in the 2022 working code but not in `Counterfactual Replication.R`. The published country rows, in which Switzerland and Belgium show public surpluses of exactly −4,358,000, have not been reconciled with any surviving code.

A revised table for Ireland and Sweden. `Counterfactual Replication.R` also builds the results table a second time, with Ireland and Sweden, like Switzerland, using their own reported demolitions. Ireland reports them for 1966–98 and Sweden for 1954–79 and 1989–2019; other years are filled the same way as Switzerland's. Ireland's demolition rate rises to 0.5 to 0.8 per cent a year, and its row falls from 7,076,000 to −2,263,000. Sweden's falls to 0.04 to 0.24 per cent, and its row rises from 2,137,000 to 10,370,000. The Western European Average moves from 4,254,000 to 4,154,000, and no other row changes. Neither revision is reliable as it stands: the Irish result depends heavily on which demolition series is used, and Sweden's reported demolitions leave out most of the homes that left its stock, which its reported stock implies were several times larger. The revised table is therefore written to a separate file and does not replace the table described above.

Rounding. Any difference between the worked figures here and the results table is due to display rounding; the code rounds only its final values.

Interpolation

In the United Nations data, housing stock values are not given consistently across Western Europe before 1960, and there are gaps of a year or more for certain countries. The estimated stock S_t fills them (`code/Summary Replication.R`, which reproduces the stored series). Within each country, in date order:

1. completions and demolitions are interpolated linearly across gaps of up to four years;

2. the ratio of demolitions to completions is taken in every year that has both, and carried forward from the most recent such year, or back from the first, into years without;
3. net additions are completions multiplied by one minus that ratio, and a year's stock includes that year's net additions.

A country that never reports demolitions has no ratio, and its net additions are treated as zero. Norway is the only such country among those in the report, so its estimate is a straight line between reported stocks.

When there is a gap **between** years in reported housing stocks, the stock is rolled forward from the earlier report on net additions, and the difference between the rolled-forward value and the next report is spread evenly across the gap (linear interpolation, for gaps of up to 35 years). For example, Austria reports 2.14 million homes in 1951 and 2.37 million in 1963. Completions over 1952 to 1963 total 504,800. Austria reports demolitions only from 1961, so the 1961 ratio of 2.5 per cent is carried back to earlier years, giving 14,300 demolitions and 490,500 net additions. The reported stock rose by only 229,600, so the shortfall of 260,900 is spread evenly: 21,700 a year over the twelve years.

When there is no data **before** a given year's reported housing stock value, the stock is rolled backwards from the first report on net additions. This matters for 1955 in two countries: Germany, whose first reported stock is for 1958, and Belgium, whose first usable report is for 1963 (its 1948 figure is excluded as inconsistent with the later series). Belgium reports 3.24 million homes in 1963. In 1962, 45,800 homes were built and 2,000 demolished, a ratio of 4.4 per cent, which is carried back to earlier years. Between 1956 and 1963, 365,000 homes were built and 18,600 demolished, leaving 346,400 net additions, which implies a housing stock of 2.89 million in 1955.

For historic England and Wales estimates, the demolitions between two census counts are the difference between gross building and the change in the stock, distributed **proportionally** to the gross building in each year. For example, the stock is 3,924,000 in 1861 and 4,520,000 in 1871, a rise of 596,000. As 749,000 homes were built in 1862 to 1871, this implies 153,000 demolitions. These are distributed over 1861 to 1870, when 702,000 homes were built, so each year's demolitions are

$$\frac{\text{Annual gross building}}{702{,}000} \times 153{,}000.$$

For graphs, methodological changes mean that discontinuities can arise – for instance, some countries' data appears to alternate between including and dropping vacant dwellings. For example, after the UN data source ends in 2000, many countries switch to counting inhabited dwellings only. In certain cases, the statistical authority will provide inhabited data before 2000 as well, and this data, although not being used in the calculations, is used in the graphs to avoid a discontinuity. If this is not possible, then a consistent interpolated estimate of the housing stock using the first method is used for graphs. This smoothed series is stored separately (`X5 (Alternative)`); the calculations in this document use S_t throughout.

For tenure, the public and private shares of construction are interpolated linearly across gaps of up to five years. Beyond that, the most recent observed share is carried forward, and before the first observation the first is carried back. The recorded value for construction is then multiplied by these shares. In practice, large gaps in the tenure data mainly occur for countries that build minimal social housing.

Changes from the February 2023 annex

- **Homes per person** divides by population; the published formula divided by the change in population.
- **Timing.** Building and demolitions are measured against the end-of-year stock, as in the code. This creates the term ε in Part I, which the code removes by scaling the UK's tenure totals, and a timing component in the Stage 6 discrepancy.
- **UK worked example.** The building rates are 2.13 and 1.98 per cent; the published 1.8 and 1.6 per cent were net of demolitions. Gross building, demolitions and net building by tenure are recomputed, and 58.6 per cent of demolitions, not two thirds, fall on private-built homes. UK tenure additions are 8.05 and 4.18 million after scaling.
- **Population growth** is measured relative to the current year's population. The Netherlands' 1955 ratio is 1/1.008, not 1/1.02.
- **Demolition adjustment.** $D_t = d_t + (\xi_t/H_t - 1)/100$. The published formula had homes per person where a stock belongs and omitted the division by 100. Negative values for Ireland are noted.
- **Stages 5–7** follow the code: gross building on the end-of-year counterfactual stock; the discrepancy defined as $J_N - (\nu_T - \sigma_0)$; and the remainder Δ allocated by the comparator's private share of gross building, against the UK's unscaled net building. The Netherlands example is recomputed. Its published gross building (23 million, of which 13.6 million private) matches the first-round stock ξ_t rather than ν_t.
- **Floor.** The published assumption that public additions stay positive is not applied by the code and does not bind. The note on the published Table 3 is new.
- **Interpolation.** Missing demolitions are filled with a carried ratio rather than by using gross completions; the Austria and Belgium examples are recomputed; Germany, as well as Belgium, is rolled back to 1955; Norway is a straight line; the England and Wales example uses the recorded 702,000 and 153,000; tenure gaps of up to five years, not four, are interpolated.
- **Tenure shares** that do not sum to one, for Finland and Switzerland, are noted.
- **Revised table.** The note on the revised table, in which Ireland and Sweden use their own demolitions, is new.

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